GUIDELINES ON HOW TO USE AI TOOLS FOR YOUR THESIS

Using AI is ***never allowed*** *as a* ***replacement for your human innovative thinking*** *or reliance on AI in the* ***absence of your critical thinking****. However, you are* ***allowed*** *to take advantage of its usefulness* ***as an assistant****. This use entails applications for sparking creative project ideas, optimizing study design, organizing data and data analysis, and synthesizing academic writing (taken from Golan et al. 2023).*

Please note that the following overview is only a guideline ***and does not cover all cases****. Please be cautious with regard to the previous sentences and question your use of AI when writing your thesis. In case of doubt, please contact your supervisor.*

YES	***Spelling, grammar, improve language*** *You may use AI tools without restrictions to check and improve spelling and grammar and rephrase sentences to improve your language.* *e.g., Grammarly, QuillBot, Wordtune*
YES	***Find and understand literature*** *You may use AI tools without restrictions to find literature for your work. For example, you can ask a tool to find literature for you on a specific topic – of course, you need to check that the literature really is a fit! Or you can use an AI tool to extend forward and backward searches. You may also use an AI tool also to understand literature. You can ask the AI to summarize certain parts of the work for your understanding and explain the model to you.* *e.g., Mindgrasp, askpaper, scite, elicit, ChatPDF*
YES *but use it carefully*	***Language translation*** *You may use AI tools to translate to make research content easier to understand or to improve your language. You can ask the AI to translate a sentence from an article into your native language to make it easier to understand. You can also write a sentence in your native language first and then ask it to translate it into the desired language of their thesis. Be careful: you must make sure that the translation is correct! You are never allowed to translate a paragraph from a research paper and copy it unchanged into your thesis!* *e.g., Google Translate, DeepL*
YES *but you have to explain in an appendix which tools were used in which parts*	***Structure and drafting of content*** *You may use AI to improve your work in a dialogue with AI. This means you can create a first outline and ask an AI tool to give feedback. Or you can give your introduction or collection of keywords to the AI and ask it which aspects might be missing. Caution: Your (critical) thinking is the most important thing here! AI should enrich you and your thoughts but never replace them!* *e.g., Hesse, Jasper AI, Isaac Editor*
YES *but you have to indicate in the text which tool was used*	***Visualize data, write code, write models*** *You may use AI to create code for smaller tasks and improve your code for larger tasks. For example, you can give AI a selection of your research data and generate Python code for visualization. Or, you can give your code to an AI and have it improve it. You can also have an AI help you with debugging.* *e.g., ChatGPT x VSCode, Github Copilot, Obviously*
NO	***Letting AI take the lead*** *You are never allowed to use AI tools to have your work (or parts of your work) completely written by AI. Thought leadership and content must always come from you and may only be enhanced with AI tools as in the cases mentioned above.*

"When submitting a piece of work ask yourself this question […] "how did you put together this piece of work?" would the answer clearly and convincingly convey that the work is substantively yours in both content and expression?

If the process you went through to arrive at the final product doesn't have your intellect, time, blood, sweat and tears firmly and clearly at the centre of it then you have probably crossed from legitimate to not legitimate."

(CANTA, Issue 5, 2023)